TRINETRA IoMT

Next-Generation Intrusion Detection & Zero-Trust Autonomous Defense System for Internet of Medical Things

Version: 2.4.0 (Pure ML Edition)	**Date:** September 2026	**Status:** Production / Validated
Environment: Fast-API + Next.js 16 + React Native	**Dataset:** CIC-IoMT-2024 (UNB)	**AI Models:** Isolation Forest & Random Forest

1. Executive Summary & Problem Statement

The **Internet of Medical Things (IoMT)** connects life-critical medical instruments—including bedside electrocardiogram (ECG) monitors, smart infusion pumps, implantable pacemakers, and pulse oximeters—to smart hospital networks. While this hyper-connectivity enhances clinical diagnostics, it introduces severe vulnerabilities: medical devices frequently rely on legacy unencrypted protocols (e.g., raw MQTT, unauthenticated TCP), exposing them to in-transit telemetry falsification, Denial of Service (DoS) floods, and medication overdose attacks.

Unlike corporate IT systems where cyber threats compromise data confidentiality or cause financial loss, attacks on IoMT directly threaten **human life**. Falsified biometrics can deceive ICU doctors into missing acute cardiac arrest, or prompt lethal drug over-infusions. **TRINETRA** solves this via a dual-tier AI intrusion detection system (IDS) coupled with an autonomous Zero-Trust Intrusion Prevention System (IPS) that detects, analyzes, and isolates threats in sub-millisecond time.

> ***Key Healthcare Security Dilemma:*** *Traditional corporate firewalls (Snort, Zeek) only inspect IP/TCP packet headers. They cannot detect clinical data poisoning where an adversary transmits valid MQTT frames with lethal dosage commands (500 mL/h). TRINETRA bridges this gap with simultaneous physiological and network anomaly fusion.*

2. System & Software Architecture

TRINETRA is structured as a 5-tier decoupled micro-architecture following the classical engineering paradigm: **Sensors / Input Data** → **Ingestion & Pre-Processing** → **Dual-Tier AI IDS** → **Zero-Trust Prevention** → **Presentation & Outputs**.

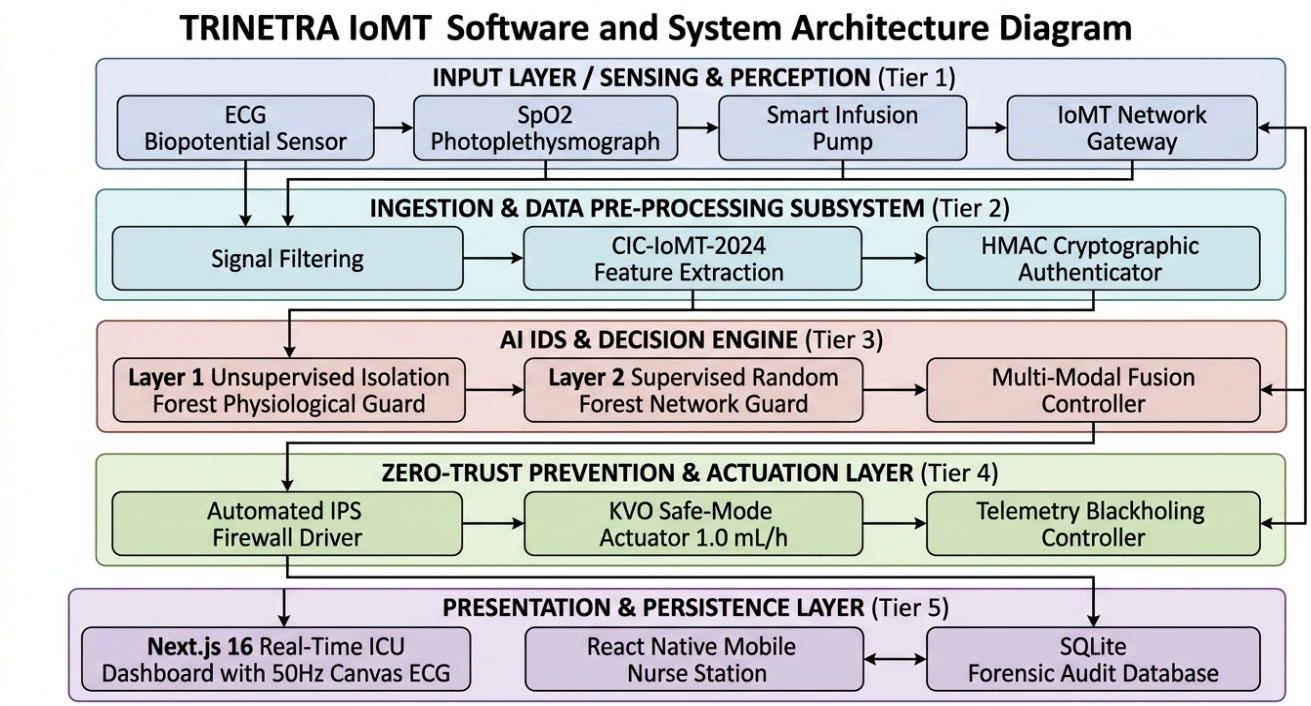


Figure 2.1: Formal 5-Tier Software and System Architecture of TRINETRA IoMT

Tier Layer	Key Modules	Signal / Data In	Action / Output
Tier 1: Perception	ECG, SpO2, Pump, Pacemaker	Biopotentials (mV), Optical absorption	Digitized telemetry & MQTT frames
Tier 2: Ingestion	Feature Extractor, HMAC Checker	Raw JSON, packet timestamps	5 CIC-IoMT flow features + HMAC state
Tier 3: AI Engine	Isolation Forest & Random Forest	Biometrics + 5 Flow features	Anomaly Score & Threat Probability
Tier 4: Prevention	Autonomous IPS, KVO Actuator	Unified Threat Score (> 0.80)	MAC Quarantine, 1.0 mL/h KVO Fallback
Tier 5: Presentation	Next.js 16 Web & Mobile App	Verified telemetry & Alarm signals	50Hz Canvas ECG, Push Alarms, DB Log

3. Machine Learning Engine & Mathematical Formulation

TRINETRA decouples defense into two dedicated machine learning models operating in parallel:

3.1 Layer 1: Physiological Guard (Unsupervised Isolation Forest)

Monitors clinical plausibility without requiring prior attack labels. It evaluates 4 biometric inputs: Heart Rate (bpm), SpO2 (%), Lead Impedance (Ω), and Infusion Flow Rate (mL/h).

Mathematical Formula: Anomaly score $s(x, n) = 2^{-E(h(x)) / c(n)}$, where $E(h(x))$ is average path length across isolation trees and $c(n)$ is the average path length of unsuccessful searches in a Binary Search Tree. Values $s \geq 0.6$ indicate acute physiological anomalies.

3.2 Layer 2: Cyber Network Guard (Supervised Random Forest)

Trained on 100,000+ flow records from the benchmark **CIC-IoMT-2024** dataset across 5 statistical network features:

Feature Name	Header	Description	Normal Range	Attack State
Duration	Duration	Flow session duration (ms)	1.0 - 5.0	Micro-bursts or long holds
Packet Rate	Rate	Transmission frequency (pkts/s)	0.5 - 5.0	Surges > 80 - 250
Packet Size	Tot size	Cumulative byte volume	120 - 500	Massive volumetric floods
AVG Size	AVG	Mean packet payload bytes	128 - 256	Fragmented frame spikes
IAT	IAT	Inter-Arrival Time (ms)	200 - 1000	Drops < 5 ms in floods

4. Attack Vectors & Autonomous Mitigation Matrix

TRINETRA implements zero-trust graduated responses based on attack vector severity:

Vector	Severity	Target Device	Attack Mechanism	Autonomous IPS Mitigation
Override	**Critical**	Infusion Pump	Lethal overdose command (500 mL/h) & rapid pacing (180 ppm)	**KVO Safe Fallback (1.0 mL/h)** + MAC Quarantine
Drain	**High**	Battery Subsystem	Continuous high-draw power cycling draining 5% battery/s	Power-Throttle Activated & Subsystem Isolated
Fault	**Warning**	ECG Monitor	Loose lead with 10,000 Ω contact impedance spike	Nurse Station Lead Inspection Alert Dispatched
DoS	**Critical**	Network Gateway	High-rate packet flood (> 200 pkts/s) dropping alarms	Rate-Limiter Active & Source IP Blackholed
MitM	**Critical**	Pulse Oximeter	Forged vitals with invalid HMAC-SHA256 signatures	Corrupted Packets Dropped + Port Quarantine
Replay	**High**	Bedside Monitor	Looping recorded healthy vitals to mask cardiac arrest	Timestamp Expiry & CUSUM Zero-Variance Drop

5. Academic Viva & Project Defense Master Guide

Q1: What is the core novelty of TRINETRA over standard hospital security?

Answer: Standard IT firewalls (Snort, Zeek) only inspect packet headers and cannot detect clinical data poisoning where an attacker sends legitimate MQTT frames with lethal dosage rates. TRINETRA uses a dual-tier AI architecture evaluating both cyber flow features and physiological plausibility, coupled with an automated actuator override (KVO mode) to safeguard patient life.

Q2: Why is Isolation Forest used for Layer 1 instead of a supervised classifier?

Answer: Every patient has a unique baseline (e.g., resting heart rate varies from 50 to 100 bpm). Supervised classifiers require pre-labeled attack datasets for every clinical scenario. Isolation Forest is unsupervised; it models the patient's multidimensional normal space and isolates anomalous deviations without prior labels.

Q3: How does the system prevent vein thrombosis during an infusion pump attack?

Answer: Instead of shutting down the pump completely (which causes intravenous blood clotting and vein collapse), TRINETRA's Zero-Trust IPS actuates a Keep-Vein-Open (KVO) safe mode delivering a harmless 1.0 mL/h flow rate to preserve vascular access.

Q4: What dataset was used for training, and why was it chosen?

Answer: Trained on CIC-IoMT-2024 (Canadian Institute for Cybersecurity, UNB). It is the 2024 healthcare benchmark testbed featuring 40+ hardware and simulated medical devices operating across MQTT, BLE, Zigbee, and Wi-Fi.

Q5: How does the 50Hz ECG generator achieve smooth animation without lag?

Answer: It computes an analytical Fourier-Gaussian PQRST mathematical formula streamed at 50Hz over WebSockets and rendered onto an HTML5 Canvas using double buffering, completely eliminating DOM overhead.